LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 6

Group Number: sampling, quantized, encoding

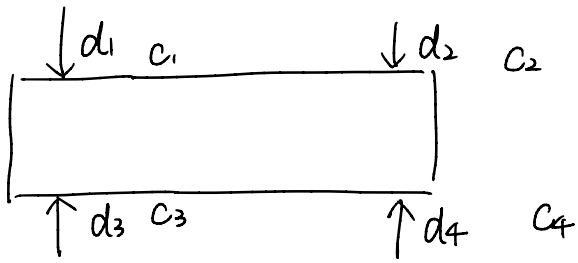
Name:

called digitalization.

1. Suppose we have an audio signal. How to convert it into a digital audio signal?
   We use Pulse code modulater for conversion.
2. Once analogue audio is converted into digital audio, how to transmit it over the air? We use analogue modulation like ASK FSK PSK or QAM.
3. We want to digitize the human voice containing frequency from 0 to 4000 Hz. What is the bit rate, assuming 8 bits per sample?
   r=8 Sampling rate = 8000 Baud
   ∴ N = 6.4×10^4 bps
4. Why do we need multiple access protocols?
   When multiple users are connected to one link, multiple access protocols is used for coordination to the link
5. What are channelization (channel partitioning) protocols?
   Is a kind of multiple protocols that available bandwidth is shared in frequency, time, and code
6. Explain the following terms: FDMA, TDMA, CDMA.

7. Assume that there are four stations in total, connected to the same channel. The data from station 1 are d1, from station 2 are d2, and so on. The code assigned to the first station is c1, to the second is c2, and so on. Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations. You can use the example given in the lecture to explain.

Frequency of the available bandwidth is divided in different parts to assign to different stations.
Time is divided in different parts to assign to different stations.
The frequency and time of available bandwidth remain the same,
each station can access the link through orthogonal code.
CDMA has only one channel occupied the whole bandwidth



take d_1 for example:

$$c_1 (c_1 \cdot d_1 + c_2 \cdot d_2 + c_3 \cdot d_3 + c_4 \cdot d_4)$$

$$= 4d_1$$

$$\frac{4d_1}{4} = d_1$$